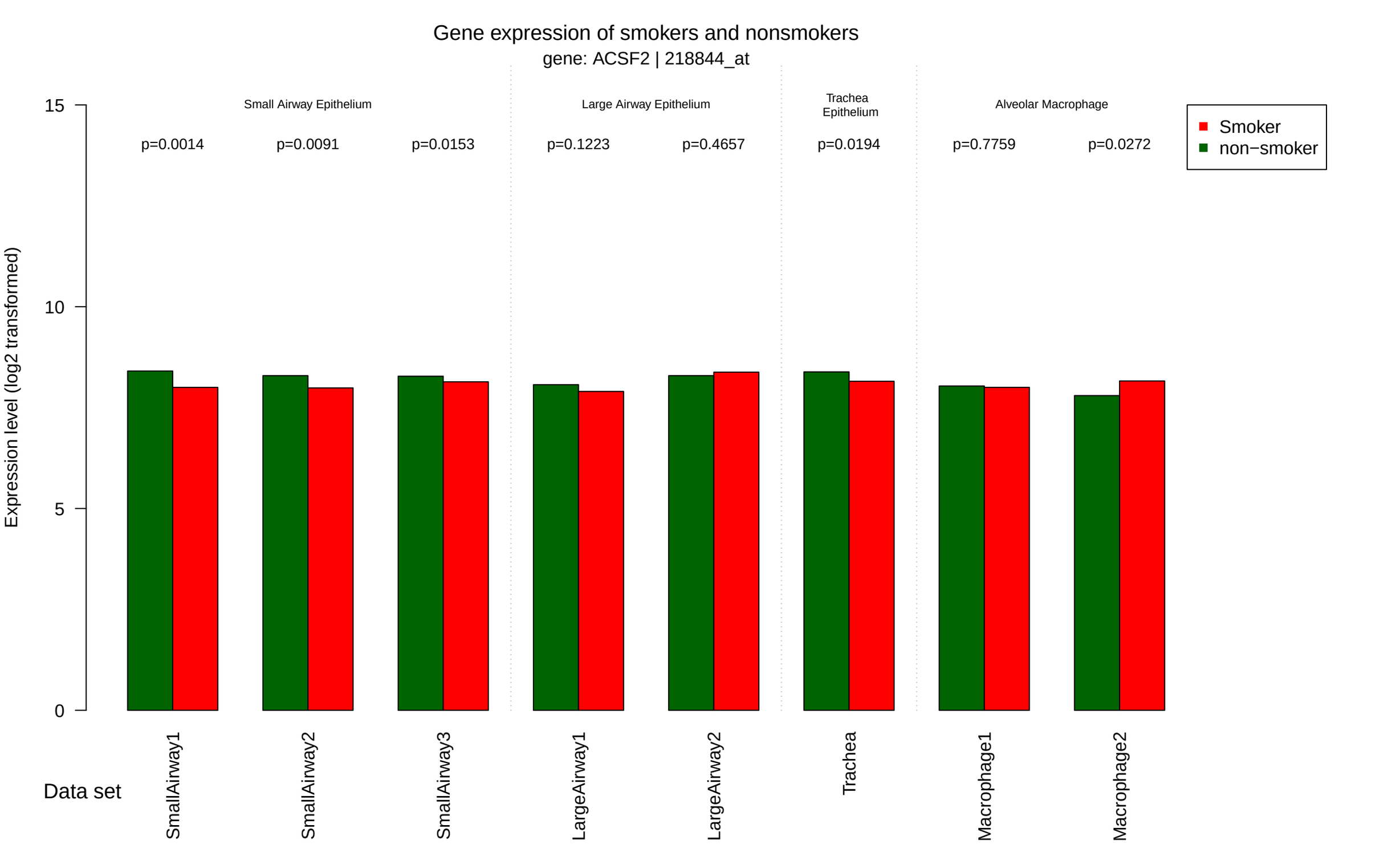
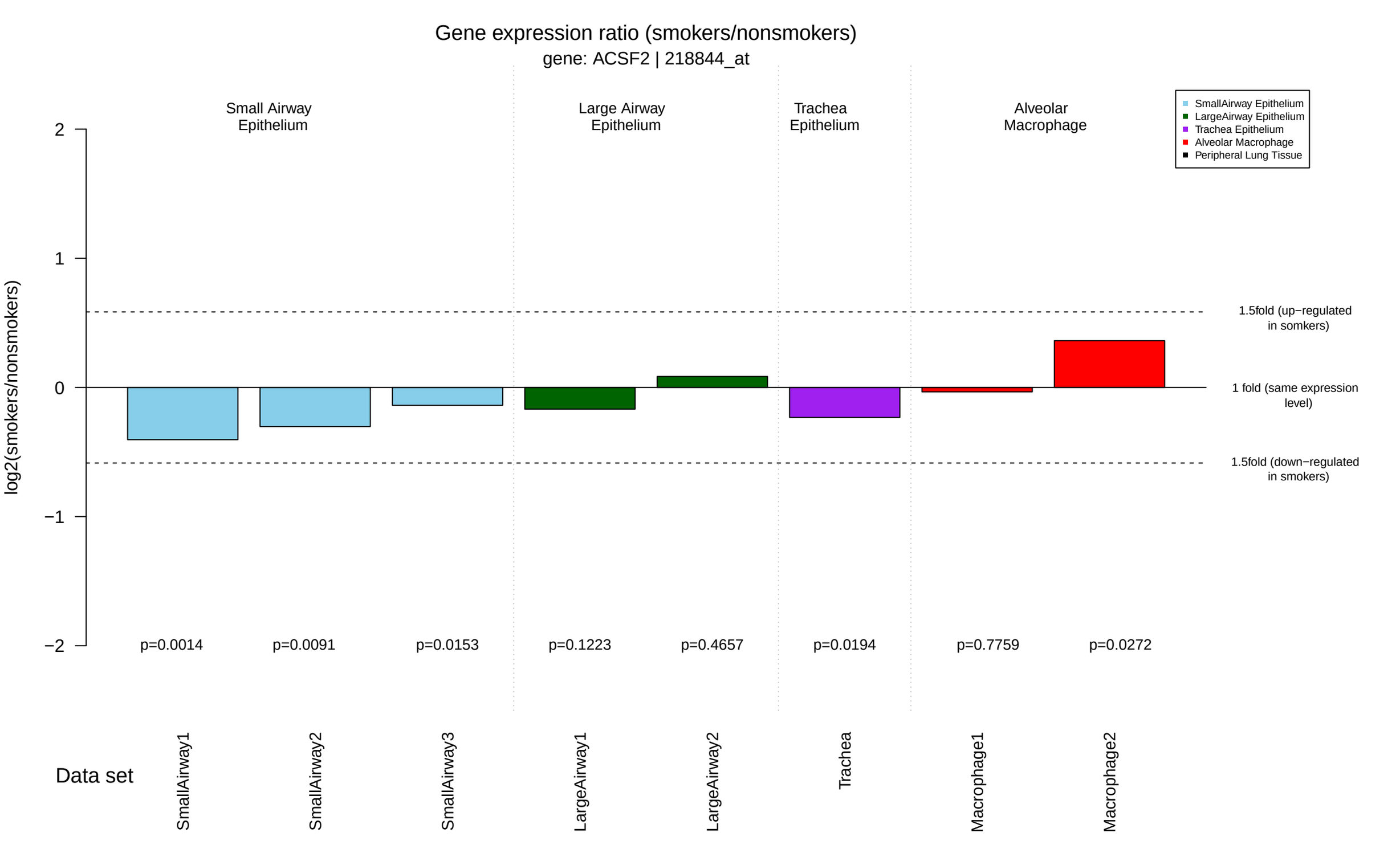
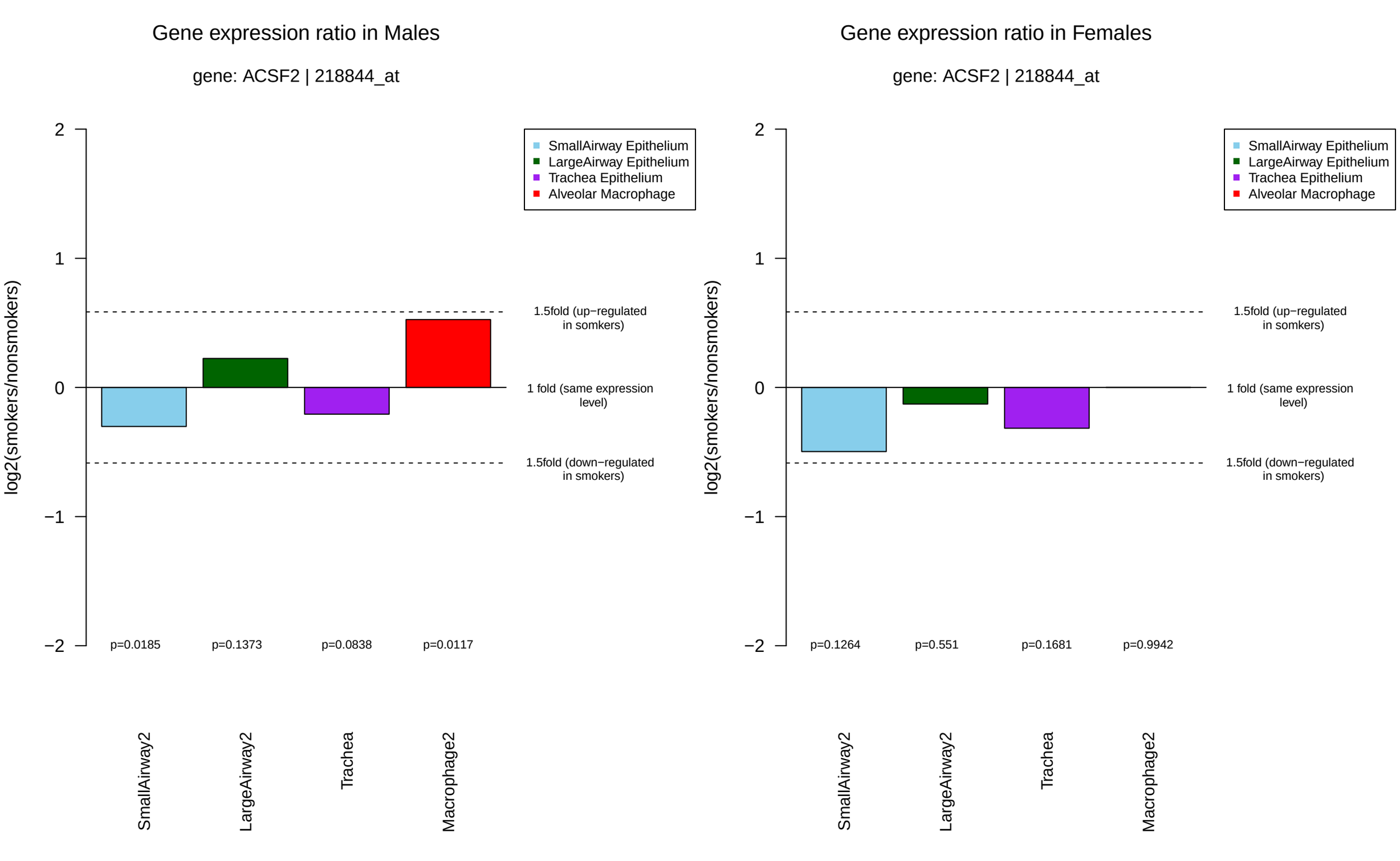
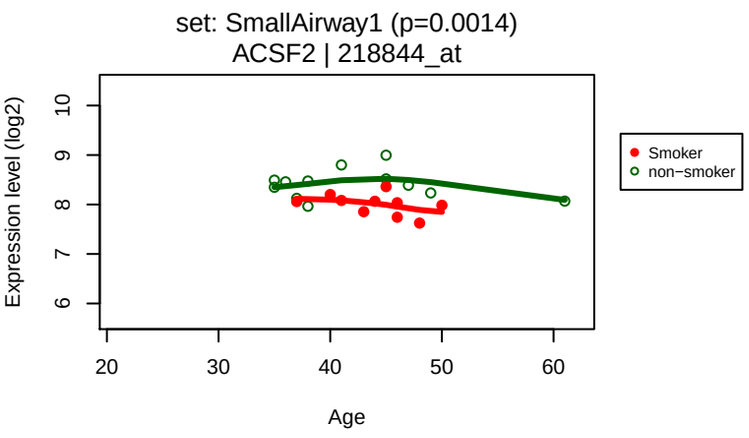
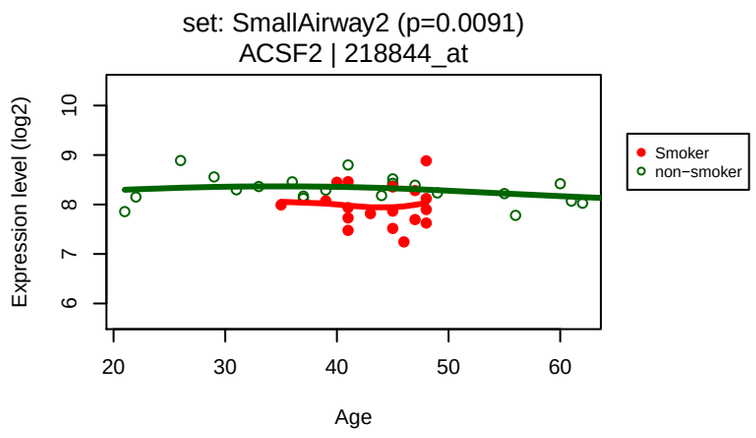
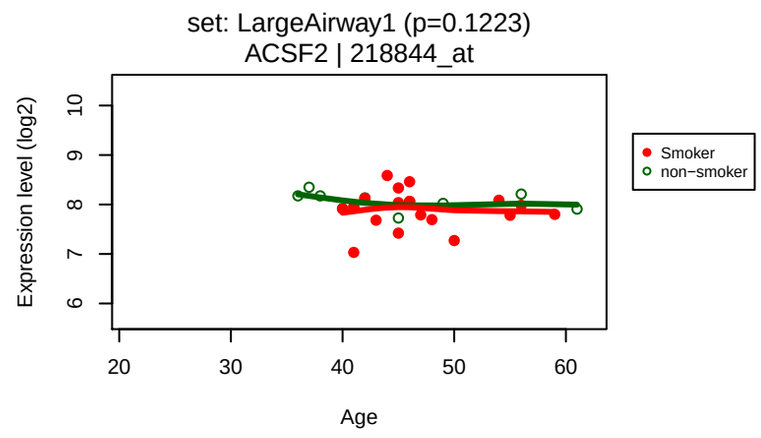

Gene expression of smokers and nonsmokers
gene: ACSF2 | 218844_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0014
p=0.0091
p=0.0153
p=0.1223
p=0.4657
p=0.0194
p=0.7759
p=0.0272
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

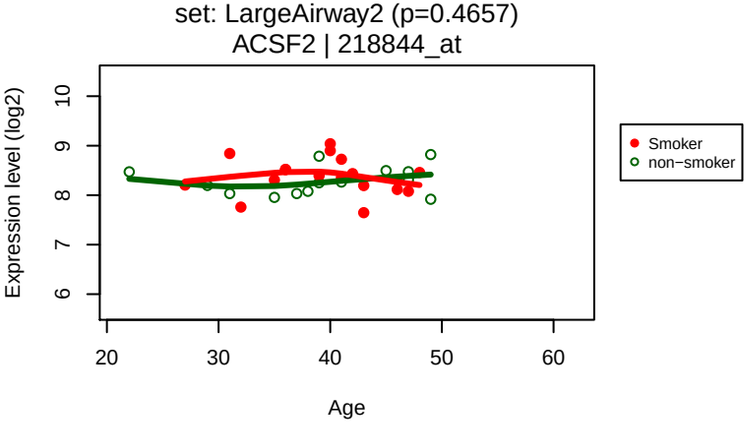

Gene expression ratio (smokers/nonsmokers)
gene: ACSF2 | 218844_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0014
p=0.0091
p=0.0153
p=0.1223
p=0.4657
p=0.0194
p=0.7759
p=0.0272
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

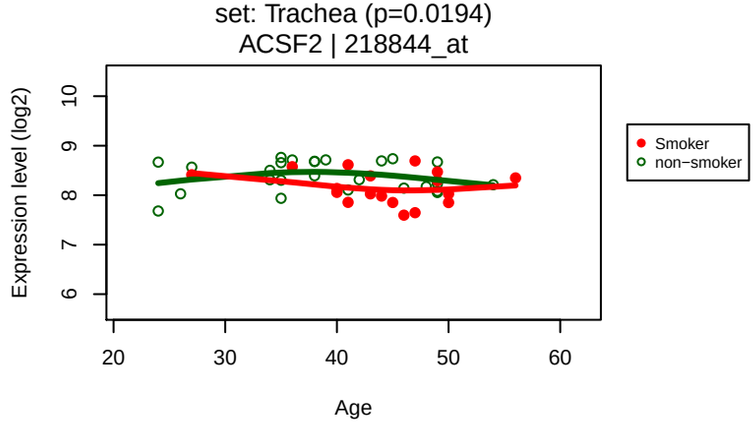

Gene expression ratio in Males
gene: ACSF2 | 218844_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0185
p=0.1373
p=0.0838
p=0.0117
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACSF2 | 218844_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1264
p=0.551
p=0.1681
p=0.9942
SmallAirway2
LargeAirway2
Trachea
Macrophage2

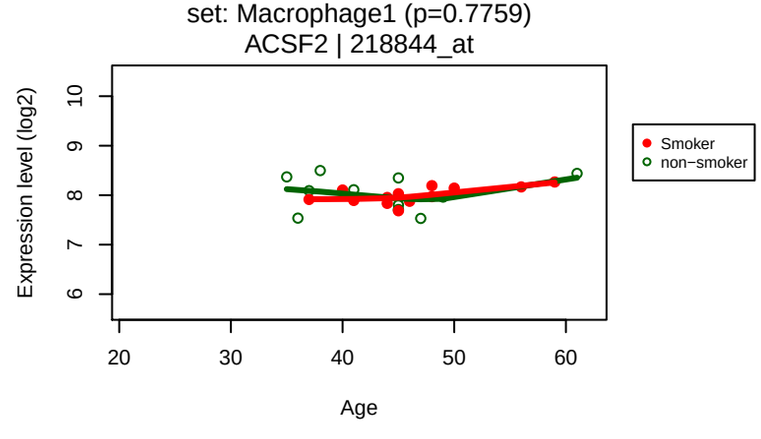

set: SmallAirway1 (p=0.0014)
ACSF2 | 218844_at
Expression level (log2)
Age
Smoker
non−smoker

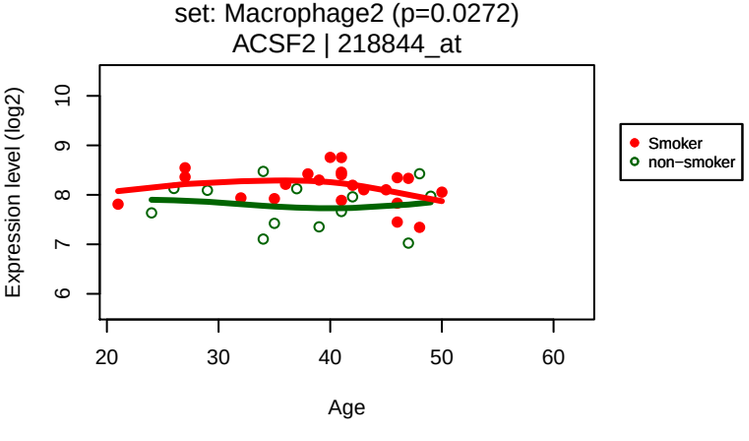

set: SmallAirway2 (p=0.0091)
ACSF2 | 218844_at
Expression level (log2)
Age
Smoker
non−smoker

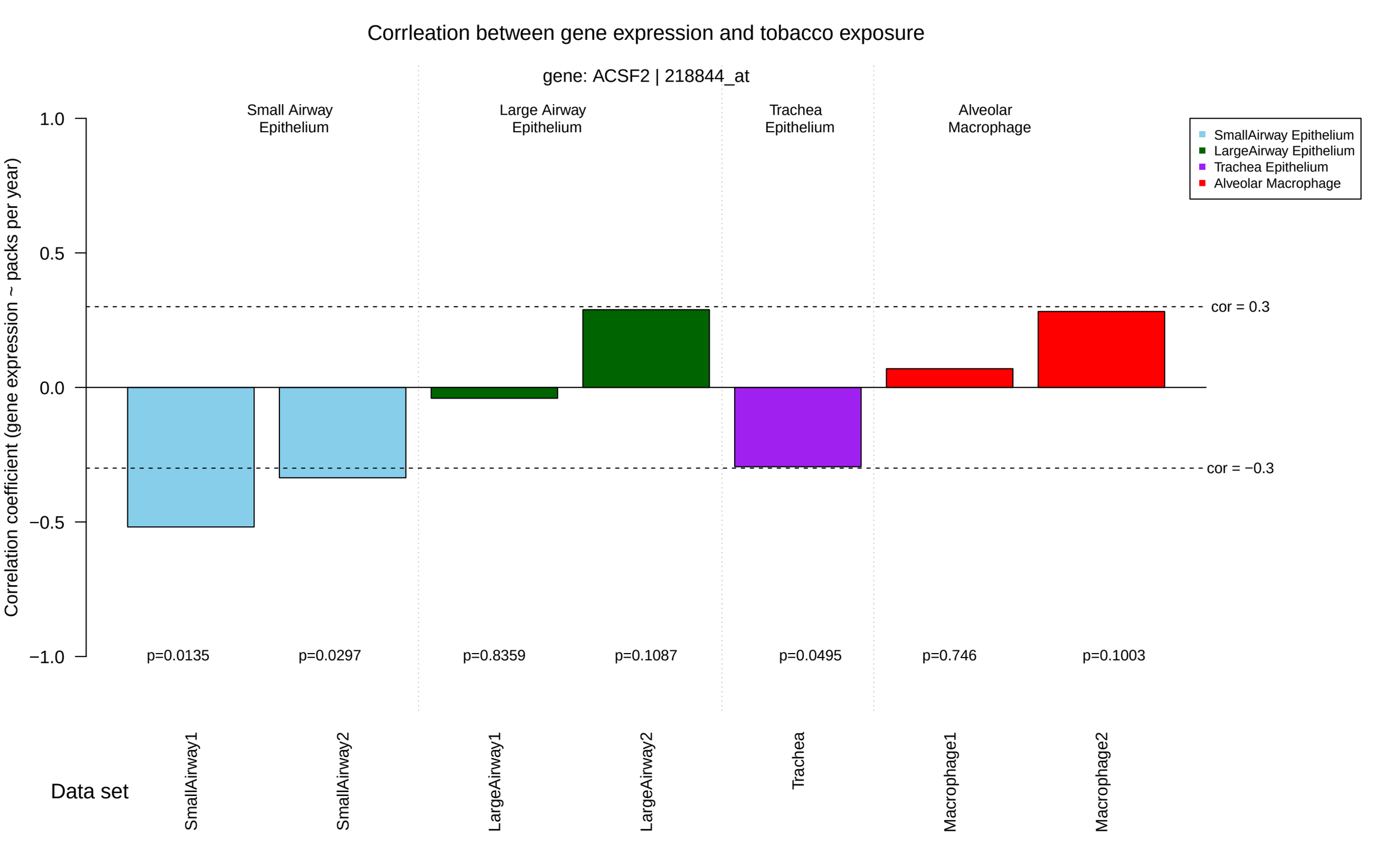

set: LargeAirway1 (p=0.1223)
ACSF2 | 218844_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4657)
ACSF2 | 218844_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0194)
ACSF2 | 218844_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7759)
ACSF2 | 218844_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0272)
ACSF2 | 218844_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACSF2 | 218844_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0135
p=0.0297
p=0.8359
p=0.1087
p=0.0495
p=0.746
p=0.1003
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2